Automatic **Landmark-Based Segmentation** of Human Subcortical Structures in MRI

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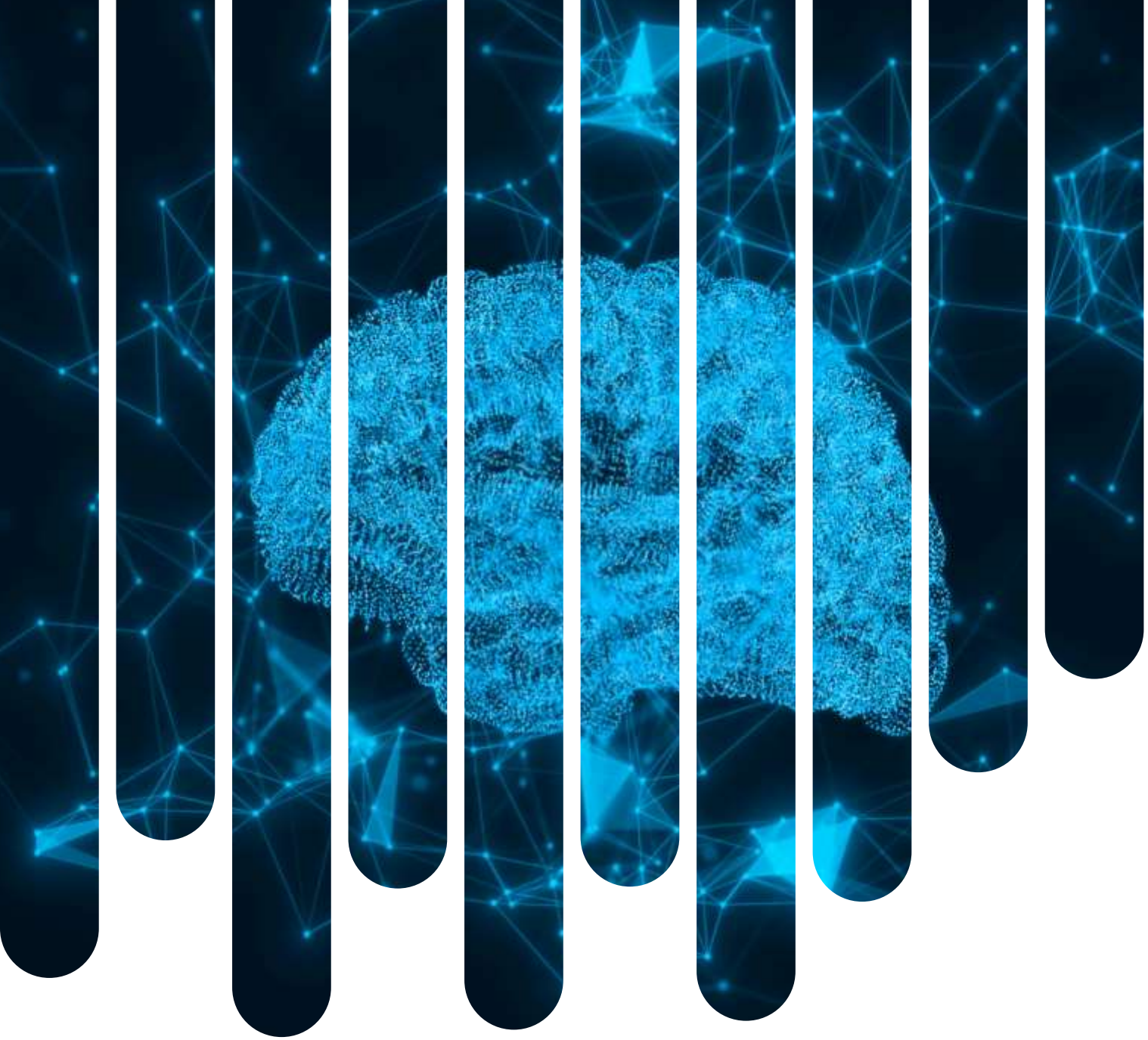

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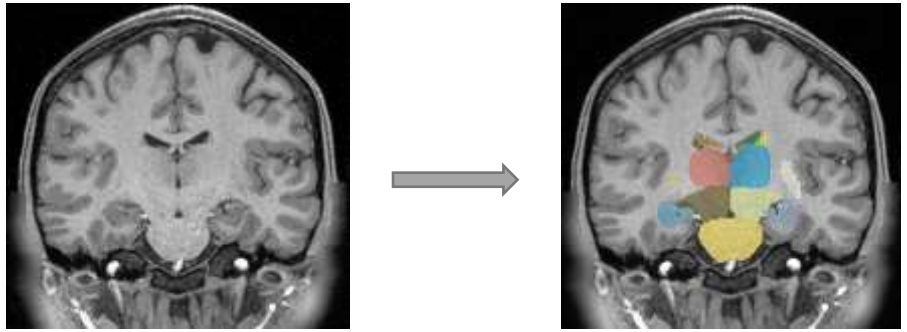
I. Neuroanatomical Context & Motivation

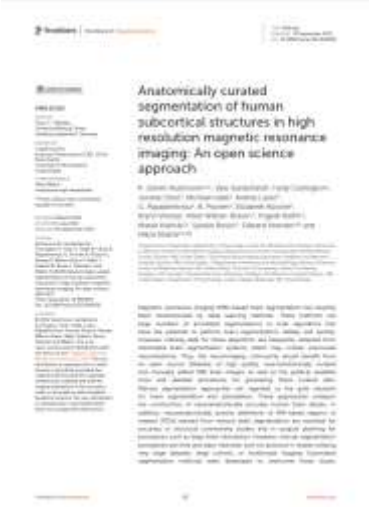
- Accurate segmentation of subcortical structures in MRI is essential for neuroimaging and clinical assessment.
- **Manual** segmentation is the **gold standard** for brain segmentation and parcellation, but it is time-consuming and hard to scale.



Harvard–Oxford Atlas 2.0 (**HOA2.0**) manual segmentation protocol

- Manual annotation follows neuroanatomical **protocols** to ensure **consistency** across experts and datasets.
- A neuroanatomical protocol defines standardized operational rules to delineate brain structures.

- 26 subcortical structures
- Landmarks + plane rules



HOA2.0 Subcortical Brain Structure Segmentation Manual

Codifying the HOA Protocol

- Protocol expressed as simple plane-based rules; Deterministic reconstruction possible
- Landmarks = anatomical reference points (*Start / end / separation*)
- 12 fused regions → 26 subcortical regions

TABLE I
SUMMARY OF THE 16 ANATOMICAL LANDMARKS

Region	Landmark(s)	ID
Putamen	First anterior appearance (L/R)	1, 2
Third ventricle	First anterior appearance	3
Nucleus accumbens–Putamen interface	Anterior contact (L/R)	4, 5
Nucleus accumbens–Putamen interface	Posterior contact (L/R)	6, 7
Nubbins	Last anterior appearance of nucleus accumbens (L/R)	8, 9
Commissures	Anterior / Posterior commissure	10, 11
Mammillary bodies	Anterior / Posterior Ventral Diencephalon (L/R)	12, 13
Continuity of atrium	First posterior appearance of inferior horn (L/R)	14, 15
Brainstem limit	Prepontine fissure	16

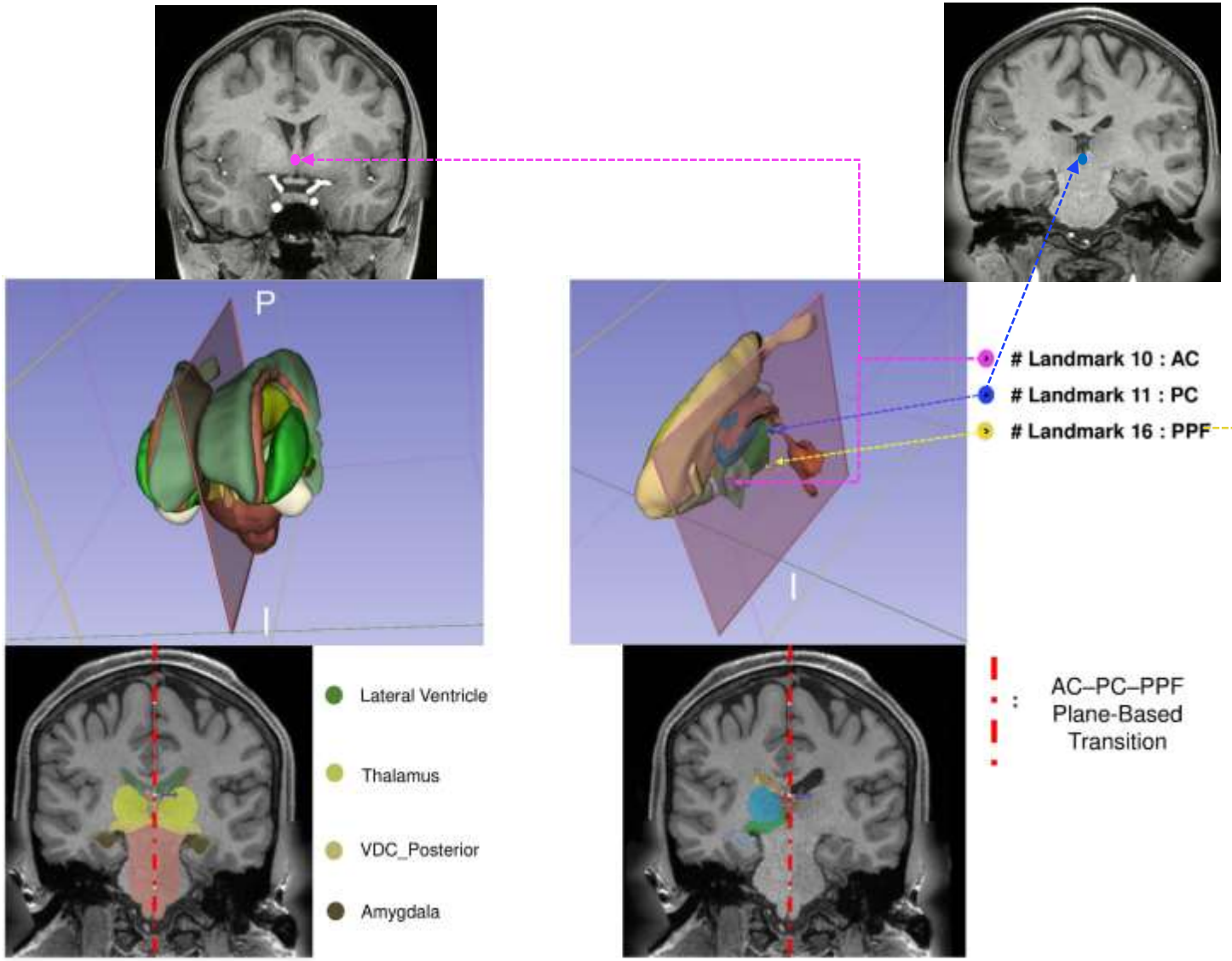
TABLE II
MAPPING FROM 26 TO 12 FUSED ANATOMICAL LABELS

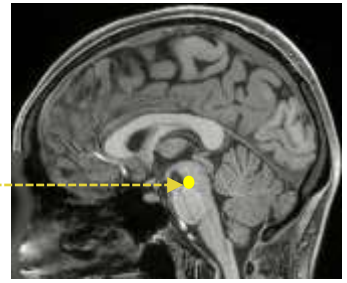
Anatomical Structures (IDs; Abbreviations)	Fused Label
(1, 2) Lateral Ventricle (LV_L / LV_R); (17, 18) Inferior Horn (IH_L / IH_R)	(1) LV + IH
(3) Cerebrospinal Fluid (CSF)	(2) CSF
(4) Third Ventricle (3V)	(3) 3V
(5) Fourth Ventricle (4V)	(4) 4V
(6, 7) Nucleus Accumbens (NAcc_L / NAcc_R); (10, 11) Putamen (Put_L / Put_R)	(5) NAcc + Put
(8, 9) Caudate (CAU_L / CAU_R)	(6) CAU
(12, 13) Globus Pallidus (GP_L / GP_R)	(7) GP
(14) Brainstem	(8) Brainstem
(15, 16) Thalamus (TH_L / TH_R)	(9) TH
(19, 20) Hippocampal Formation (HF_L / HF_R)	(10) HF
(21, 22) Amygdala (AMY_L / AMY_R)	(11) AMY
(23–26) Ventral Diencephalon (VDC_A_L/R, VDC_P_L/R)	(12) VDC

- Landmarks #4–#7 define NAcc–Putamen interface
- Clean separation despite similar intensity

HOA2.0 Subcortical Brain Structure Segmentation Manual

Example 1 : Midline splitting





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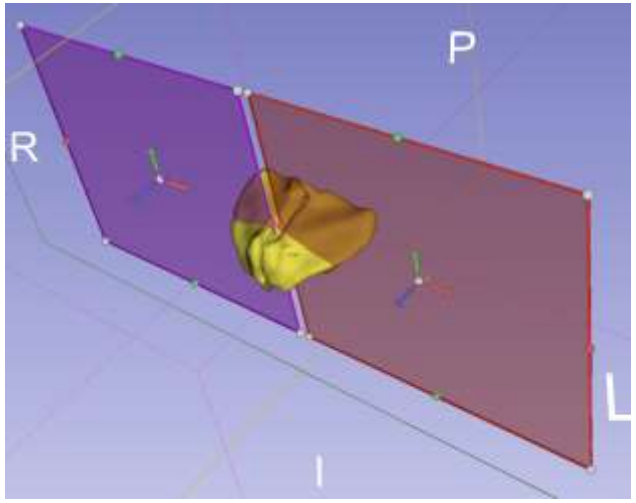
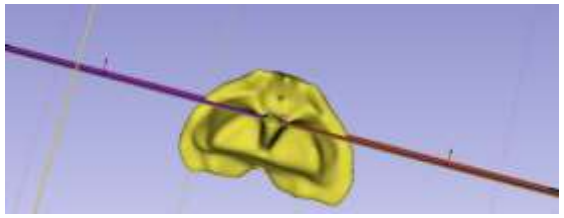
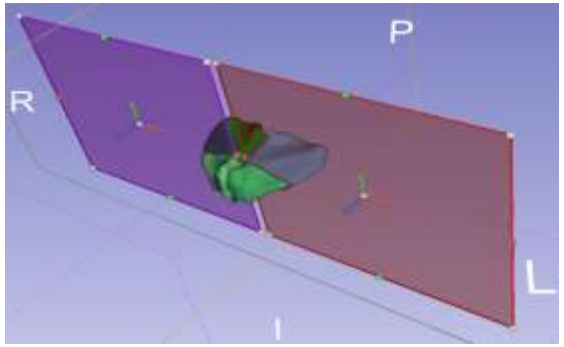
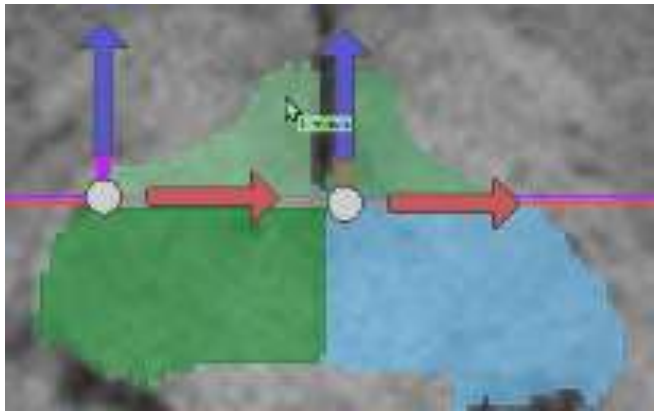
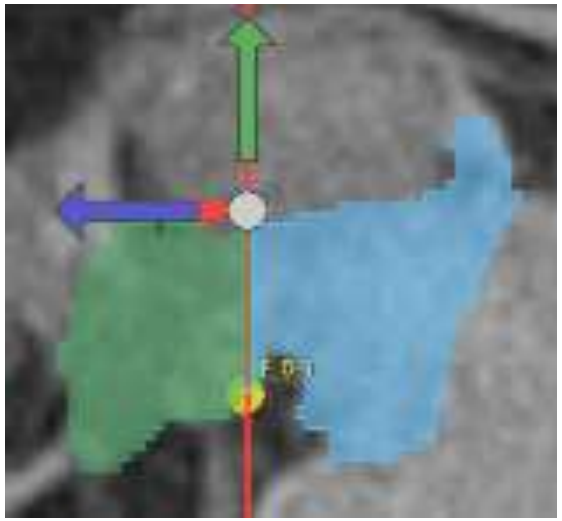
Example 2 : VDC Anterior–Posterior Separation

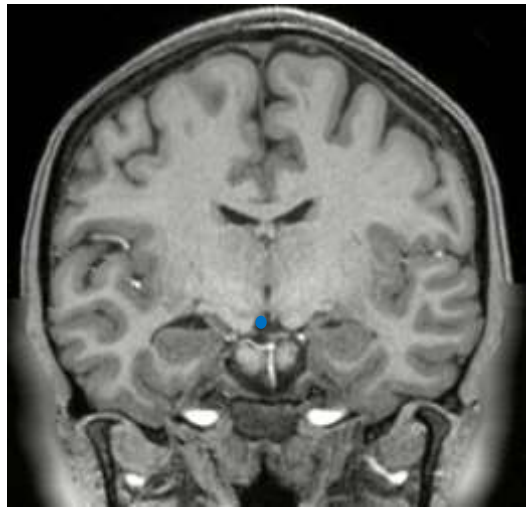




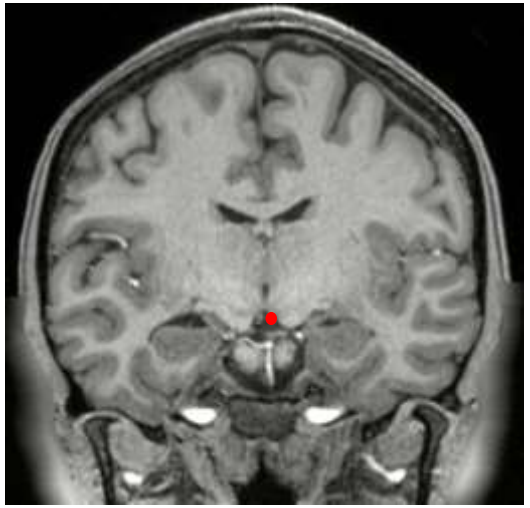








Landmark 13



Landmark 12

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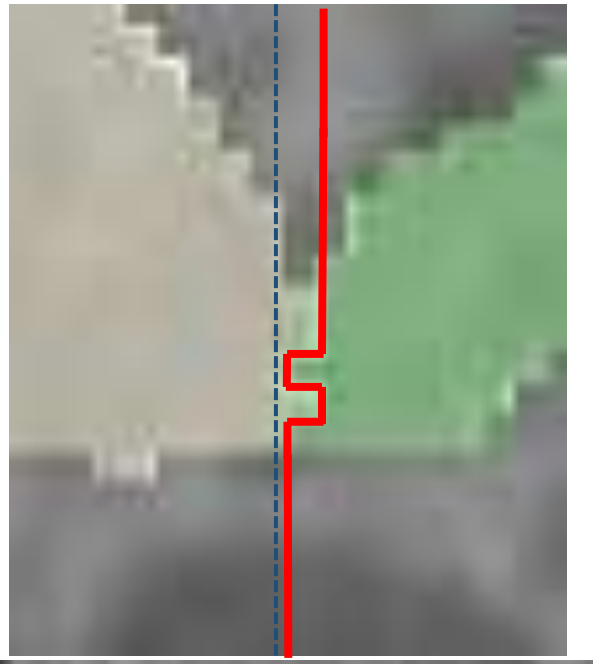
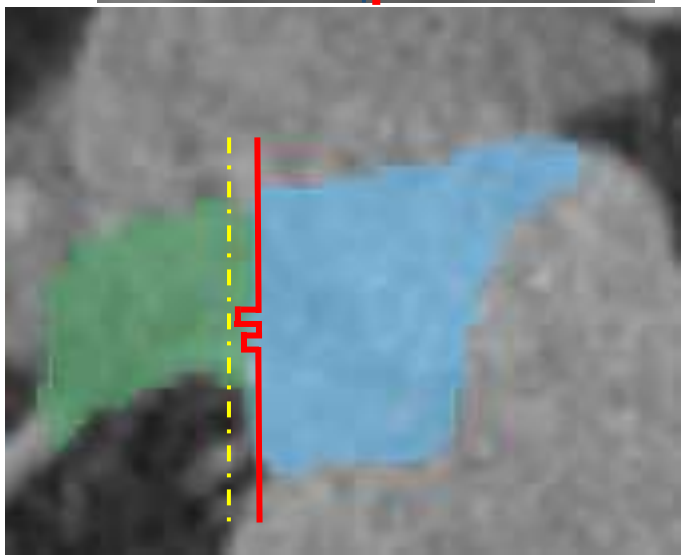
Limitation of standard deep learning

- Voxel-wise learning
- Leakage / fused structures
- No anatomical rules
- No protocol awareness

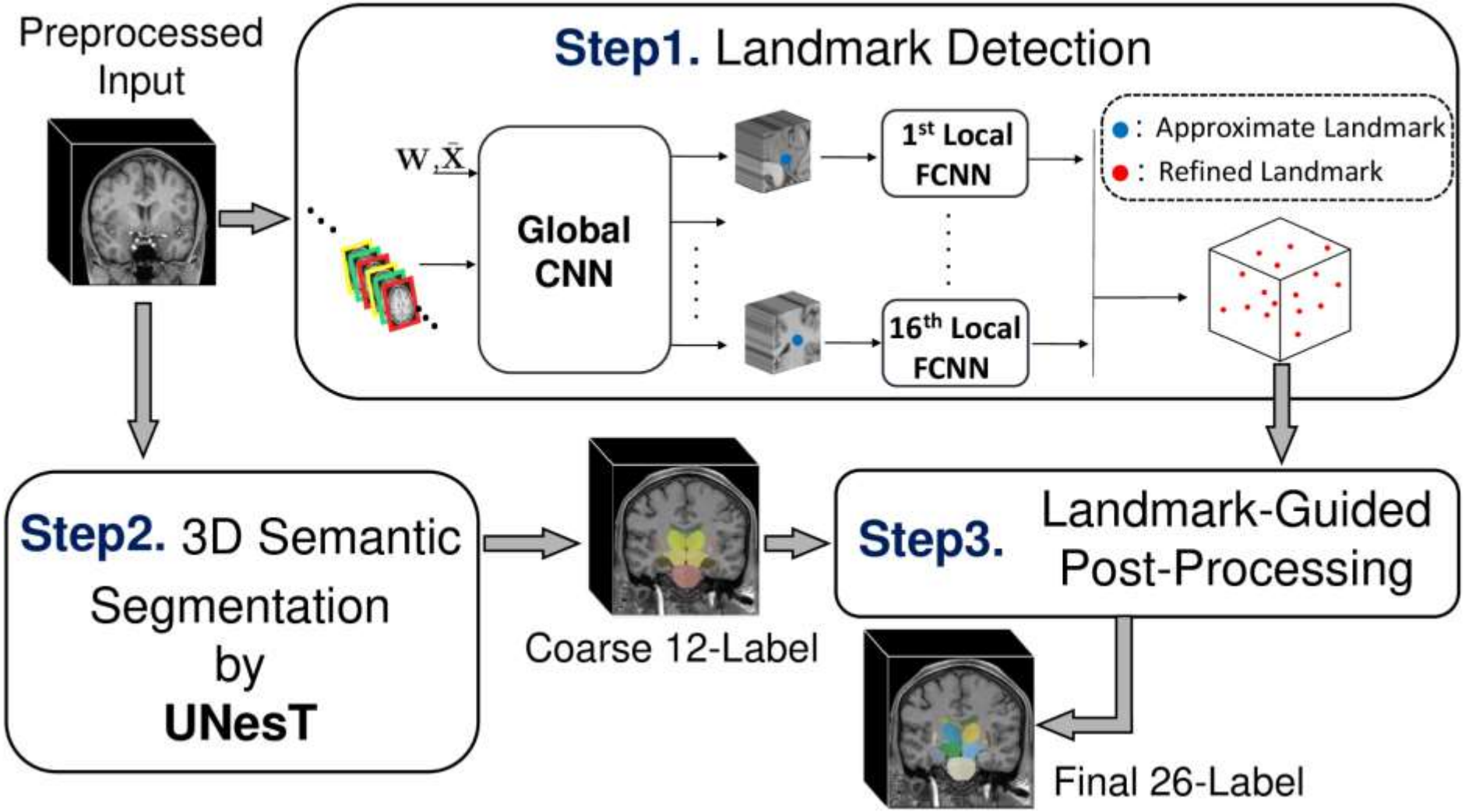


Our Goal : Towards Protocol-Compliant Segmentation

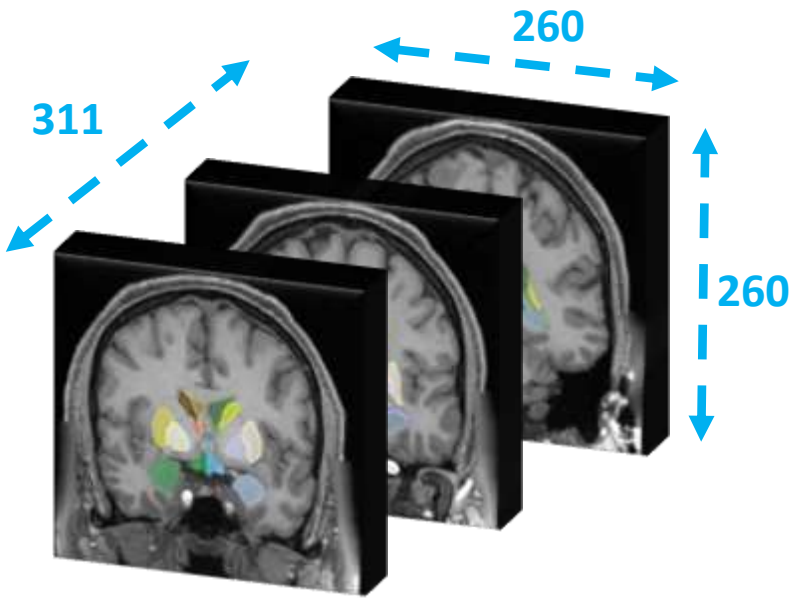
- Beyond voxel-wise accuracy
- Enforce neuroanatomical protocol rules
- Landmark-based segmentation pipeline



Overall pipeline



HOA Subcortical Dataset



DATA

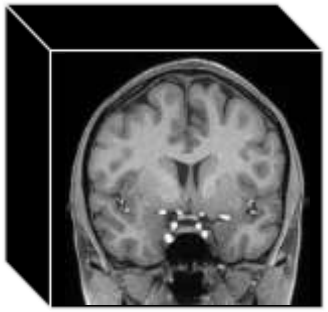
- 100 T1-weighted MRI (HCP Young Adult)
- 26 subcortical structures
- 16 anatomical landmarks (HOA protocol)

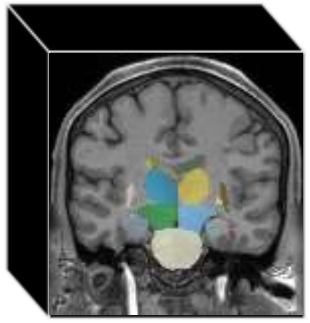
Resolution & Split

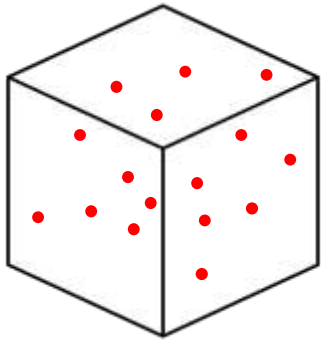
- 0.7 mm^3 — 260 × 311 × 260
- Train: 80 / Val: 10 / Test :10

Preprocessing

- Skull-stripping (SynthStrip)
- Z-score normalization





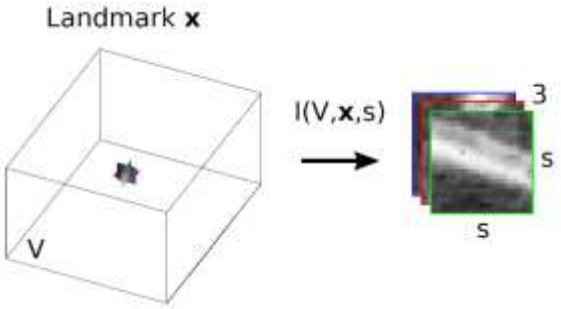


manually annotated in this study

➢ Step 1.1. Global Model – PIN

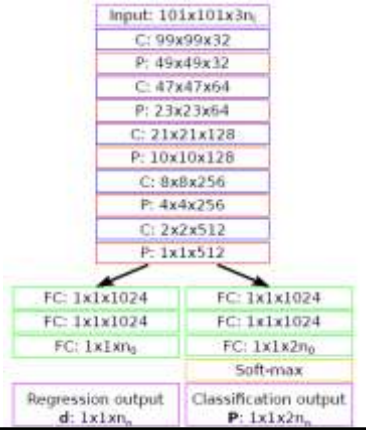
Patch-Based Iterative Network

- Initialize landmarks at their mean coordinates.
- Patch Extraction : 2.5D orthogonal views

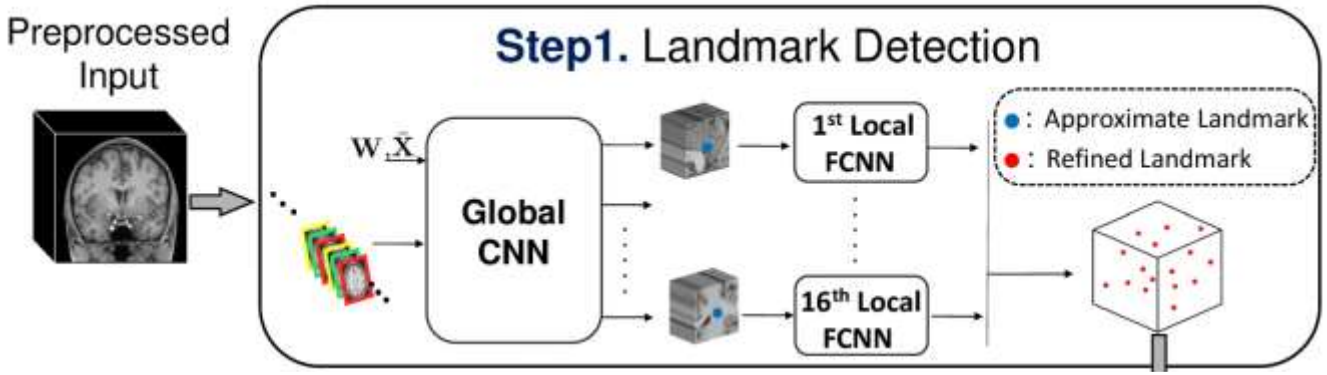


- CNN Forward Pass :

$$x_{t+1} = x_t + P_{\max} \cdot d$$



Iterative Refinement until convergence



- PCA (Principal Component Analysis) :

The network predicts updates in PCA parameter space b

$$X = \bar{X} + Wb$$

; n_b=36 < 48, retaining approximately 99.5% of the total variance

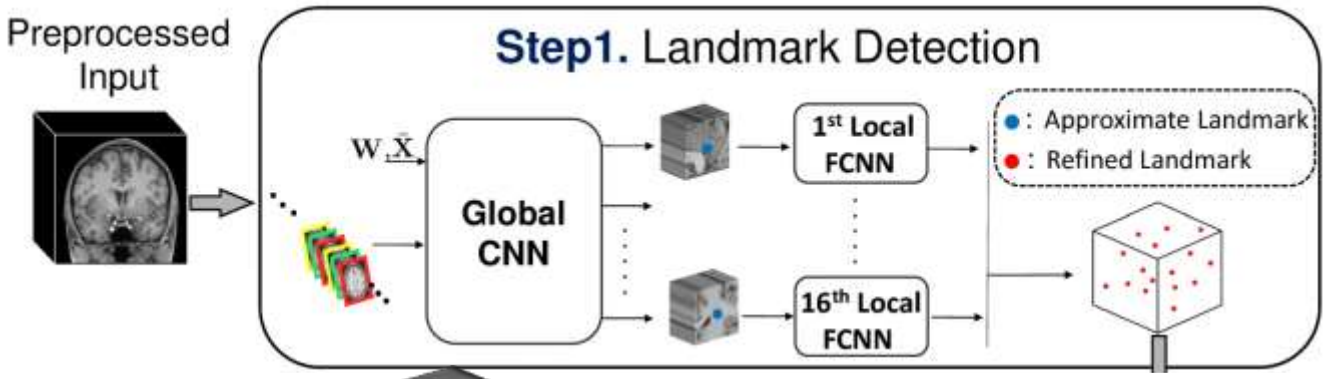
$$b_{t+1} = b_t + P_{max} \odot d \Longrightarrow X_{t+1} = \bar{X} + Wb_{t+1}$$

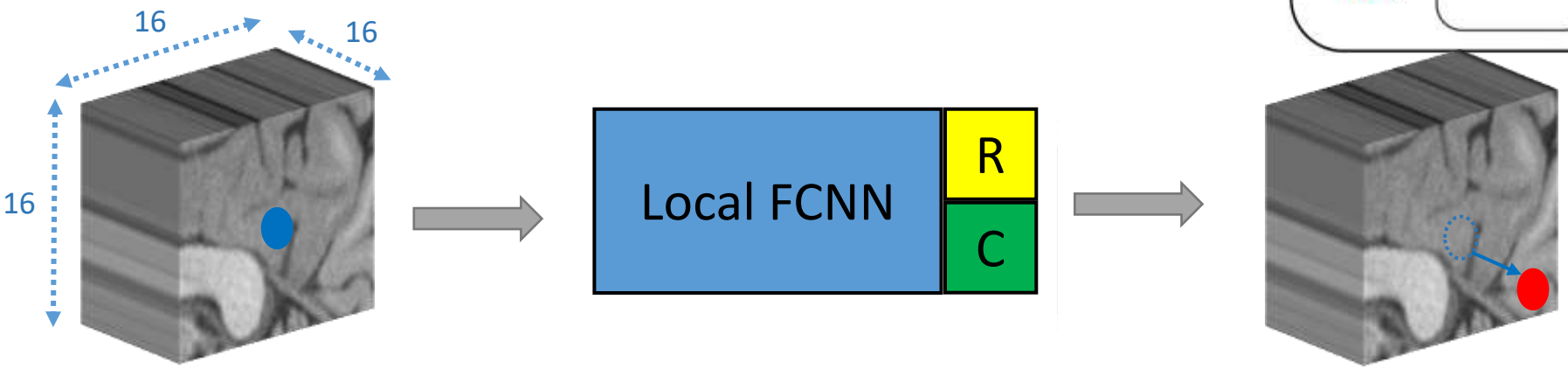
- Multi-task loss :

$$\mathcal{L} = (1-\alpha) \cdot \|d_{\mathrm{GT}} - d\|_2^2 + \alpha \cdot \mathrm{CrossEntropy}(P_{\mathrm{GT}}, P)$$

➢ Step 1.2. Local Models

refine coarse landmark predictions to achieve higher accuracy.





- Multi-task loss :

$$\mathcal{L} = \mathrm{MAE}_{\log}(\mathbf{d}, \hat{\mathbf{d}}) + \mathrm{BCE}(y, \hat{y})$$

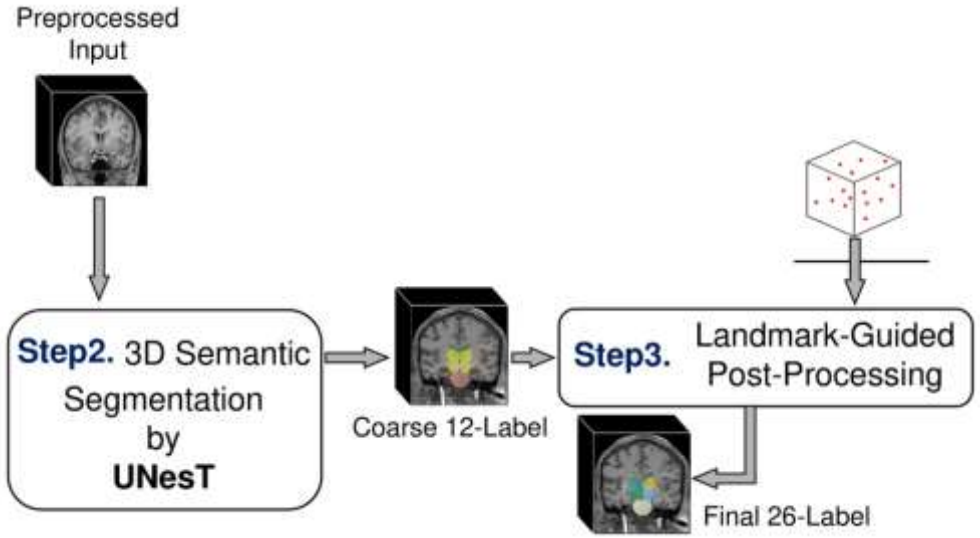
- Generating Local Training Patches
  - Global PIN error radius estimated on validation set: r_i
  - Start from ground-truth landmark position: p_i
  - Sample Gaussian displacement:

$$\boldsymbol{\delta}_i \sim \mathcal{N}(\mathbf{0}, \sigma_i^2 \mathbf{I}) \quad \text{where ;} \quad \|\boldsymbol{\delta}_i\|_2 \leq r_i \quad \text{for 95\% of samples}$$

  - Patch center computation: $\mathbf{c}_i = \mathbf{p}_i + \boldsymbol{\delta}_i$

➢ Step 2. Coarse Segmenation

- UNesT: Hierarchical 3D Transformer (State-of-the-Art)
  - Trained to segment 12 fused subcortical regions
- Training Setup
  - 300 epochs
  - AdamW optimizer
  - Dice + Cross-Entropy loss
- Baseline Model
  - Same training strategy
  - UNesT trained to directly segment 26 subcortical regions



➢ Step 3. Post-Processing

- Inputs :
  - Landmarks from Step 1
  - Coarse segmentation (12 labels) from Step 2 (UNesT)

→

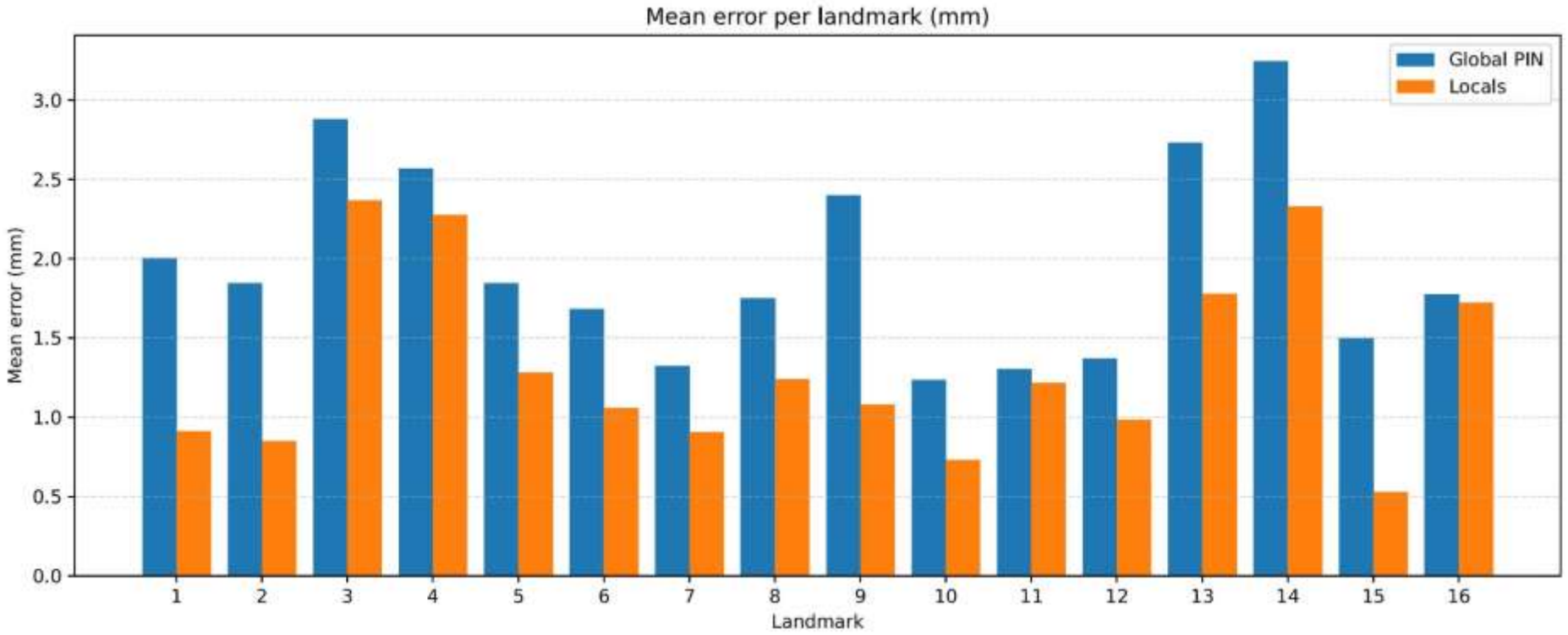
- Post-processing
  - Apply protocol plane rules
  - Landmark-guided separations

→

- Output
  - 26 protocol-compliant subcortical regions

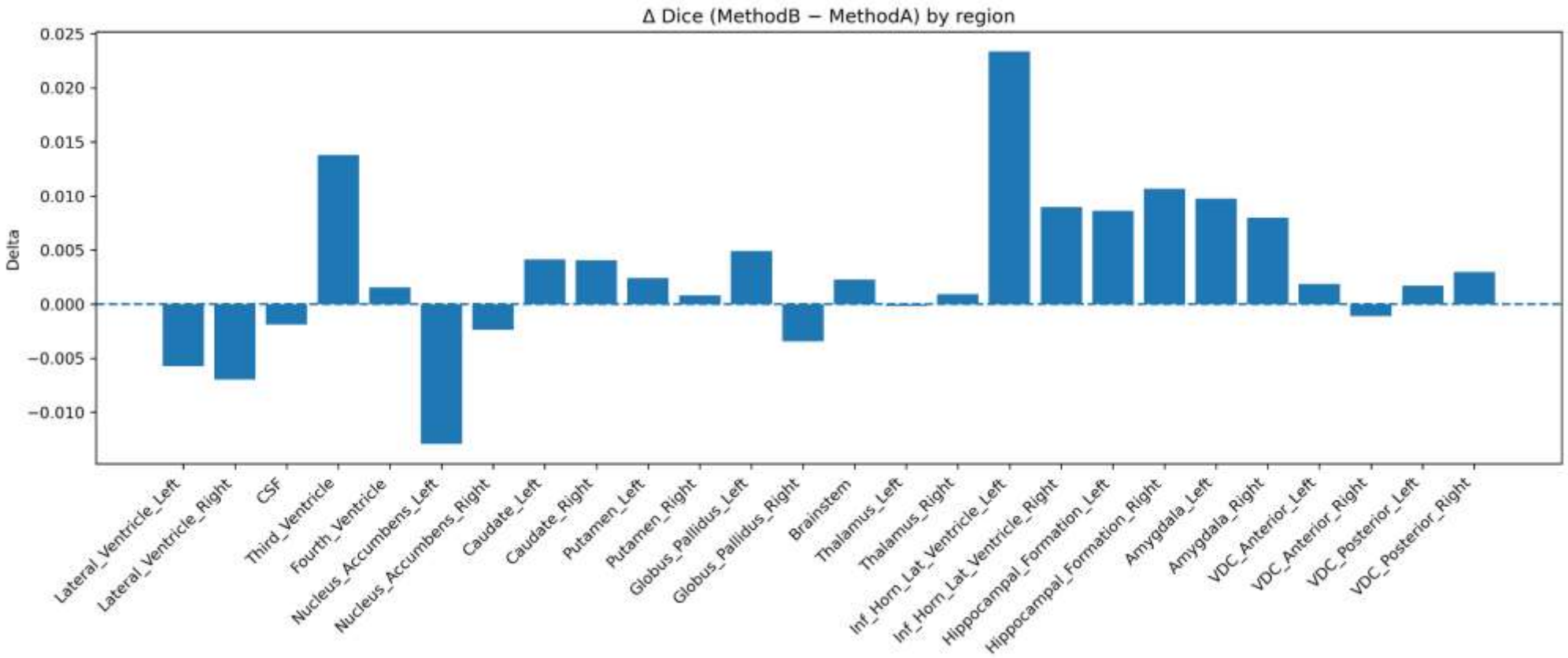
➢ **Landmark detection results :**

$$\text{Error}_i = \frac{1}{N}\sum_{s=1}^{N} \|\hat{\mathbf{p}}_{s,i} - \mathbf{p}_{s,i}\|_2$$



- Global PIN: 1.97 mm
- Global + Local: 1.33 mm
- 32% improvement

➢ Segmentation results (1/2) :



- Baseline (Method A): 0.8976
- Ours (Method B): **0.9008**
- Small global gain

➢ Segmentation results (2/2) :

$$\mathrm{PASD} = \frac{1}{\left|\mathcal{S}_{\mathrm{GT}}^{(p)}\right|} \sum_{v_i \in \mathcal{S}_{\mathrm{GT}}^{(p)}} \min_{u_j \in \mathcal{V}_{\mathrm{Pred}}^{(s)}} \|v_i - u_j\|_2 .$$

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left| y_i^{\mathrm{pred}} - y_i^{\mathrm{gt}} \right| ,$$

$$\sigma_y = \frac{1}{N} \sum_{i=1}^{N} \left(y_i^{\mathrm{pred}} - \bar{y}_{\mathrm{pred}} \right)^2 .$$

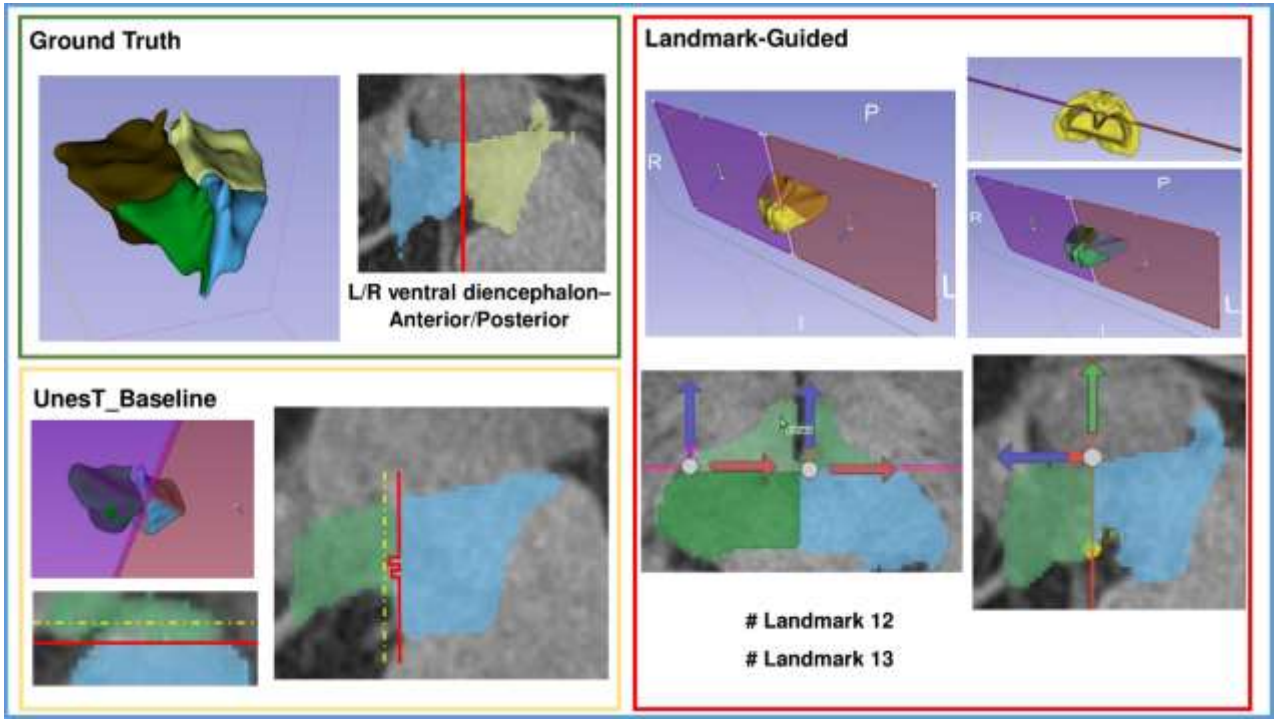
TABLE III
PASD (LOWER IS BETTER) FOR RELEVANT BOUNDARIES. A: UNEST BASELINE; B: LANDMARK-GUIDED

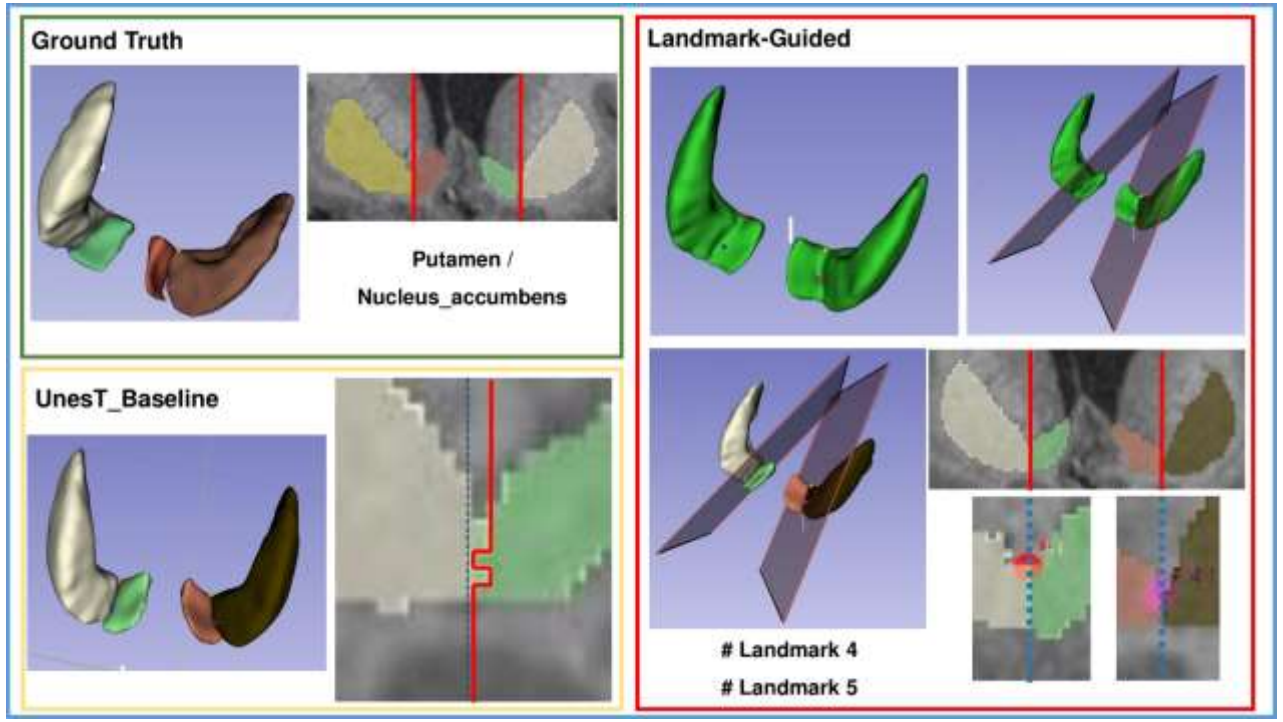
Region (Surface)	Left		Right	
	A	B	A	B
IH (posterior)	0.787	**0.524**	0.636	**0.356**
NAcc (lateral)	0.597	0.651	0.364	0.370
NAcc (posterior)	0.427	**0.388**	0.398	0.449
Put (anterior)	0.491	**0.470**	0.302	0.480
Put (lateral)	0.451	**0.424**	0.459	**0.441**
3V (anterior)	0.601	**0.514**	–	–
VDC_A (posterior)	0.650	**0.465**	0.384	**0.352**
VDC_P (anterior)	0.444	**0.413**	0.331	0.363

TABLE IV
2D METRICS OF BOUNDARY VERTICALITY AND REGULARITY. A: UNEST BASELINE; B: LANDMARK-GUIDED. LOWER IS BETTER

Region (Surface)	Left		Right	
	A	B	A	B
MAE				
IH (posterior)	0.706	0.903	0.889	0.471
VDC_P (anterior)	0.619	0.614	0.480	0.488
NAcc (lateral)	0.826	0.913	0.867	0.945
σ_y				
IH (posterior)	0.110	**0.000**	0.167	**0.049**
VDC_P (anterior)	0.153	**0.000**	0.242	**0.000**
NAcc (lateral)	0.118	**0.000**	0.095	**0.000**

➢ **Qualitative evaluation :**

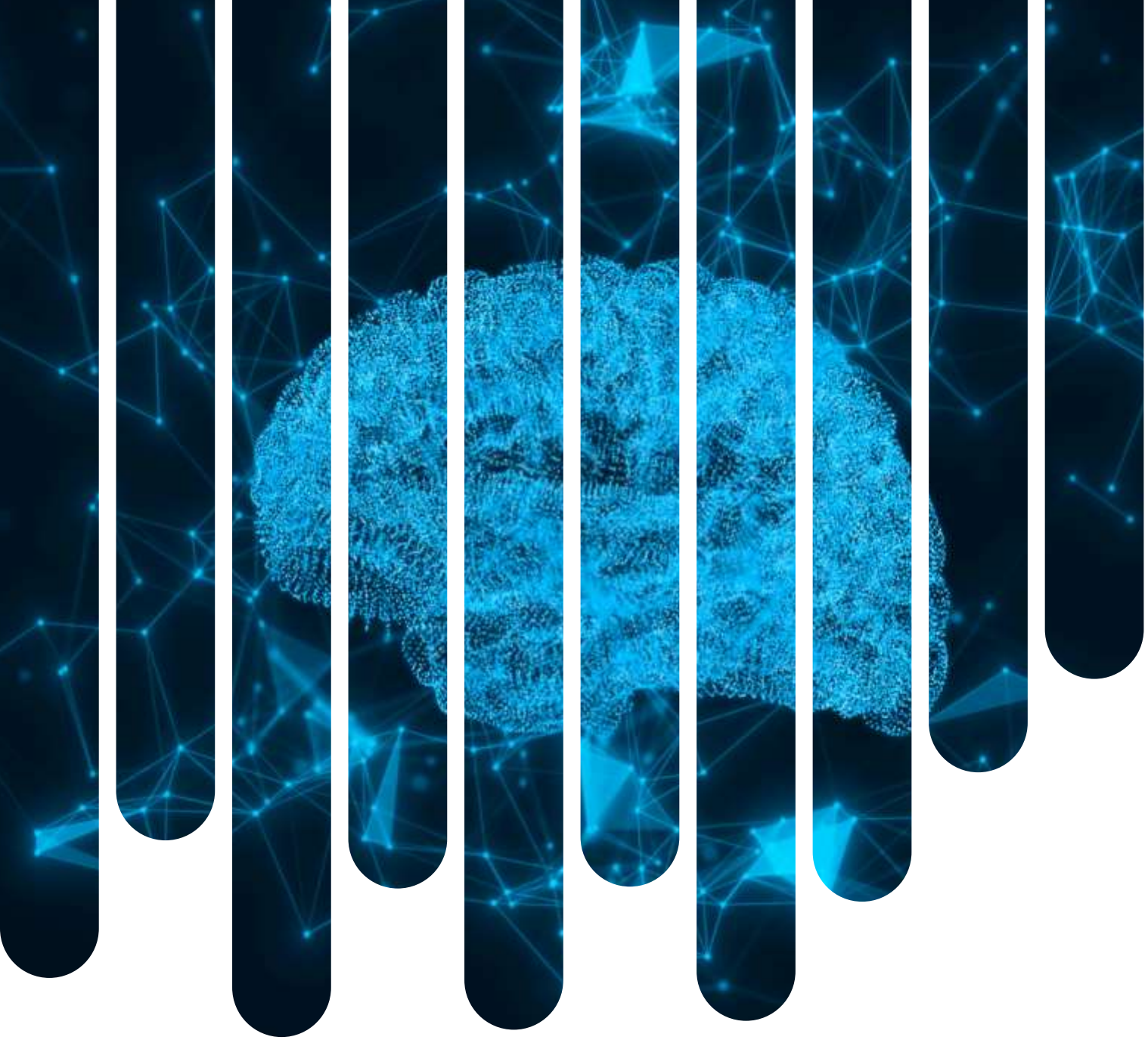




- Qualitative expert evaluation on 5 cases
- Performed by HOA co-author (R. J. Rushmore)
- Our method better follows protocol than baseline

IV. Conclusion

- Landmark-guided framework for subcortical segmentation
  - Enforces protocol-consistent boundaries
  - Modest Dice gains, strong boundary improvement
  - Expert preferred landmark-guided method
- Highlights gap between automated and manual protocols



Thank you for your attention !

Any Questions ?